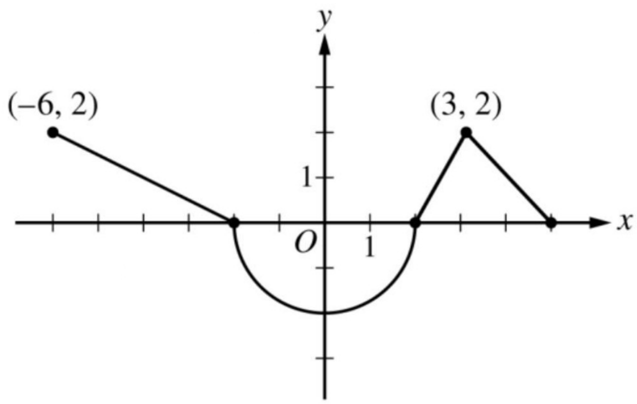
Calc Final Practice Problems
13 problems (cut down version) no calculator
answers to be released if there's time (stay tuned).

1. The graph below represents f'(x). Find f(5) given f(0)=3.



2. $3x^2 + 2xy + y^2 = 1$, find $\frac{dy}{dx}$.

3. Using the table below, if $\int_2^5 f(x)dx = 14$, find $\int_2^5 x \cdot f(x)dx$.

x	2	5
f(x)	4	7
f'(x)	2	3

4. Use differentials to approximate $\frac{1}{\sqrt[3]{28}}$

5. Kip Glazer loves to showcase her pride in MVHS by posting on Instagram. Say that she posts a selfie of her at an MVHS football game, and the function $f(t) = 100e^{-0.2t}$represents the rate of people like her post in likes per day. How many likes on her Instagram post will she have after 5 days? (Write a simplified, non decimal answer).

6. The base of a solid is bounded by the curves $y = x^2$ and $y = 4 - x^2$. Given that the cross sections of the solid perpendicular to the x-axis are triangles with the height twice the length of the base, find the volume of the solid.

7. A rectangle is to be inscribed between the parabola $y = 4 - x^2$ and the x-axis, with the base on the x-axis. What is the positive value of x that will maximize the area?

8. A ball is thrown from a 48 feet tall tower with an initial velocity of 64 feet/s. What is the acceleration of the ball after 3 seconds?

9. Using u-sub, find $\int (5x + 4)^6 (5x - 5) dx$

10. Use Simpson's rule, the integral mean-value equation, and the table below to find the average **radius** of a blood vessel.

Distance (mm)	0	60	120	180	240	300	360
Diameter(mm)	24	30	28	30	26	24	26

11. What is the total area of the region defined between the functions $x = y^3 - y$ and $x = 3y$?

12. Find the derivative of $(\cos x)^{2x+5}$, do not simplify.

13. Find $\int 1 + \left(\frac{\cos^2 x}{\sin^2 x}\right) dx$

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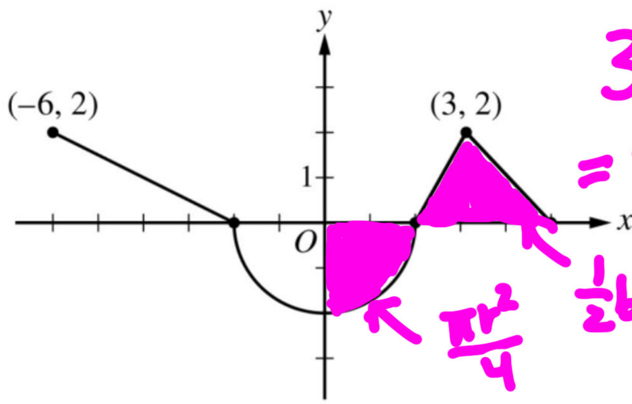
1. The graph below represents f'(x). Find f(5) given f(0)=3



$\frac{\pi r^2}{4}$ $\frac{1}{2}bh$

$3+\int_0^5 f'(x)dx$

$=3-\left(\frac{\pi 2^2}{4}\right)+\frac{1}{2}(3)(2)$

$3-\pi+3=6-\pi$

$\boxed{f(5)=6-\pi}$

2. $3x^2+2xy+y^2=1$, find $\frac{dy}{dx}$.

take derivative

$6x+(2xy'+2y)+2yy'=0$

$6x+2y+2xy'+2yy'=0$

$6x+2y+y'(2x+2y)=0$

$y'(2x+2y)=-(6x+2y)$

$y'=-\frac{(6x+2y)}{2x+2y}$

$y'=\boxed{-\frac{(3x+y)}{x+y}}$

3. Using the table below, if $\int_2^5 f(x)dx=14$, find $\int_2^5 x\cdot f(x)dx$.

x	2	5
f(x)	4	7
f'(x)	2	3

Integration by parts

$\int u\,dv = uv-\int v\,du$

$u=x$
$dv=f(x)$
$v=\int f(x$
$du=dx$

$= xf(x)-\int_2^5 f(x)dx$

$= xf(x)-14\Big]_2^5$

$(5f(5)-14)-(2f(2)-14)=\boxed{27}$

$=21-(-6)$

4. Use differentials to approximate $\frac{1}{\sqrt[3]{28}}$ → nearest value is $\frac{1}{\sqrt[3]{27}} = \frac{1}{3}$

Linearization
$f'(a)(x-a) + f(a)$

$f(x) = \frac{1}{\sqrt[3]{x}}$ or $x^{-\frac{1}{3}}$ → $f(27) = \frac{1}{3}$ → $-\frac{1}{243}(28-27) + \frac{1}{3}$

$f'(x) = -\frac{1}{3}x^{-\frac{4}{3}}$ or $-\frac{1}{3x^{\frac{4}{3}}}$ → $f'(27) = -\frac{1}{243}$ ↗ $= \boxed{\frac{1}{3} - \frac{1}{243}}$

5. Kip Glazer loves to showcase her pride in MVHS by posting on Instagram. Say that she posts a selfie of her at an MVHS football game, and the function $f(t) = 100e^{-0.2t}$ represents the rate of people like her post in likes per day. How many likes on her Instagram post will she have after 5 days? (Write a simplified, non decimal answer).

Integration as net change

$f(t) =$ rate of likes per day

$\int f(t)\,dt = \#$ of likes

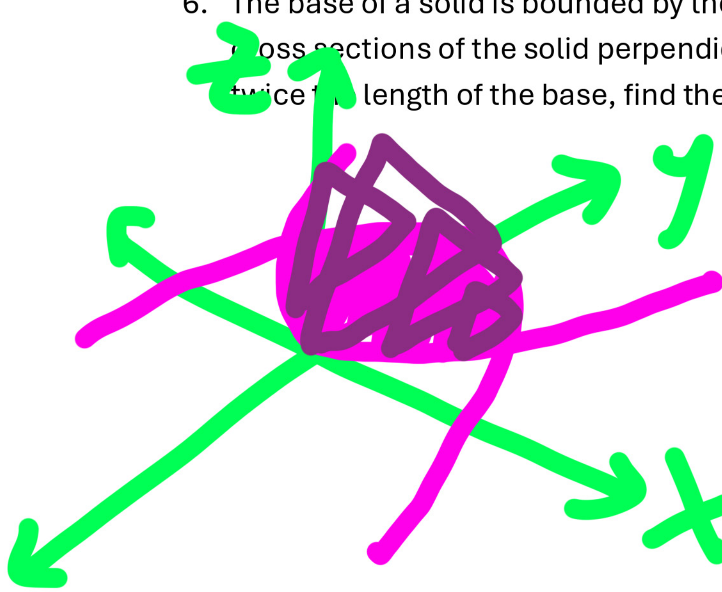
$\int_0^5 100e^{-0.2t}\,dt$

$100\int_0^5 e^{-0.2t}\,dt$

$100\left[\frac{e^{-0.2t}}{-0.2}\right]_0^5 = \left[-500e^{-0.2t}\right]_0^5 = -500e^{-1} - (-500e^0)$

$= \boxed{500 - \frac{500}{e} \text{ likes}}$

6. The base of a solid is bounded by the curves $y = x^2$ and $y = 4 - x^2$. Given that the cross sections of the solid perpendicular to the x-axis are triangles with the height twice the length of the base, find the volume of the solid.



Find Bounds
$x^2 = 4 - x^2$
$2x^2 = 4$
$x^2 = 2$
$x = \pm\sqrt{2}$

Base = Top − Bottom
$4 - x^2 - x^2$
$= 4 - 2x^2$

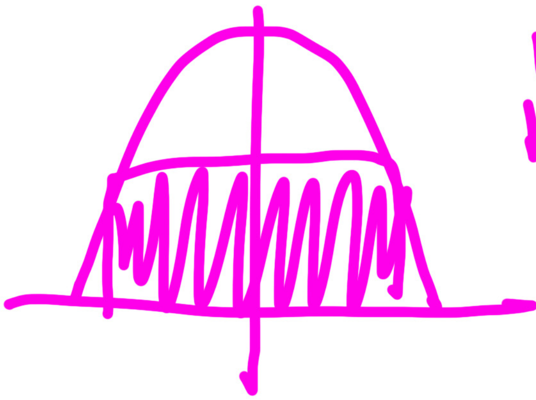
Height = 2 · base
$= 8 - 4x^2$

Area $= \frac{1}{2}bh$
$= \frac{1}{2}(4-2x^2)(8-4x^2)$
$= \frac{1}{2} \cdot 2(2-x^2)\,4(2-x^2)$
$= 4(2-x^2)^2$

Integrate
$\int_{-\sqrt{2}}^{\sqrt{2}} 4(2-x^2)^2\,dx \to 2\int_0^{\sqrt{2}} 4(2-x^2)^2\,dx$
$= 2\int_0^{\sqrt{2}} 16 - 16x^2 + 4x^4\,dx$
$2\left[16x - \frac{16x^3}{3} + \frac{4x^5}{5}\right]_0^{\sqrt{2}}$
$32\sqrt{2} - \frac{32(\sqrt{2})^3}{3} + \frac{8(\sqrt{2})^5}{5} = 32\sqrt{2} - \frac{64\sqrt{2}}{3} + \frac{32\sqrt{2}}{5}$

$\frac{480\sqrt{2}}{15} - \frac{320\sqrt{2}}{15} + \frac{96\sqrt{2}}{15}$
$= \boxed{\frac{256\sqrt{2}}{15} \text{ units}^3}$

7. A rectangle is to be inscribed between the parabola $y = 4 - x^2$ and the x-axis, with the base on the x-axis. What is the positive value of x that will maximize the area?



Width: $2X$

Height: $4 - X^2$

$\therefore$ Area $= (2x)(4-x^2)$

$= 8x - 2x^3$

Area$' = 8 - 6x^2$

$8 - 6x^2 = 0$

$8 = 6x^2$

$\frac{4}{3} = x^2$

$\boxed{X = \sqrt{\frac{4}{3}}}$

8. A ball is thrown from a 48 feet tall tower with an initial velocity of 64 feet/s. What is the acceleration of the ball after 3 seconds?

Position Function: $-16t^2 + 64t + 48$

lol I realized none of the given values are needed, but If you want to know the velocity after 3 seconds

$s'(t) = -32t + 64$

$\boxed{\text{Velocity @ 3s: } -32 \text{ ft/s} \quad \text{Acceleration @ 3s: } -32 \text{ ft/s}^2}$

9. Using u-sub, find $\int (5x+4)^6(5x-5)dx$

$\int (5x+4)^6((5x+4)-9)$

$u = 5x+4$

$du = 5dx$

$\frac{du}{5} = dx$

$\frac{1}{5}\int u^6(u-9)$

$\frac{1}{5}\int u^7 - 9u^6$

$\frac{1}{5}\left(\frac{u^8}{8} - \frac{9u^7}{7}\right) + c$

$= \frac{u^8}{40} - \frac{9u^7}{35} + c$

$\boxed{= \frac{(5x+4)^8}{40} - \frac{9(5x+4)^7}{35} + C}$

pls pardon my writing I was on trackpad

10. Use Simpson's rule, the integral mean-value equation, and the table below to find the average **radius** of a blood vessel.

0 2 3 4

Distance (mm)	0	60	120	180	240	300	360
Diameter(mm)	24	30	28	30	26	24	26

$$= \frac{b-a}{6n}\left[f(x_0)+4f(x_{\frac{1}{2}})+2f(x_1)\cdots f(x_n)\right]$$

$$= \frac{360}{6(4)}\left[24+4(30)+2(28)+4(30)+2(26)+4(24)+26\right]$$

$$= 15(494) = 7410$$

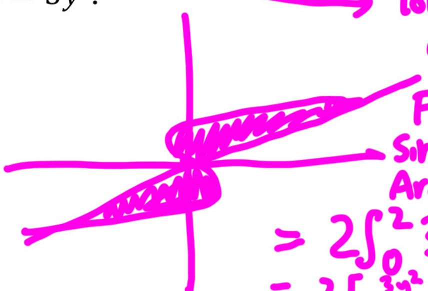
Now time to find avg radius

radius $= \frac{1}{2}$ diameter $\rightarrow \frac{1}{2}(7410$

$\frac{1}{b-a}\int_a^b f(x)\,dx$

$= \frac{1}{360}\left(\frac{3705}{2}\right) = \boxed{\frac{741}{72}\text{ mm}}$, surely

11. What is the total area of the region defined between the functions $x = y^3 - y$ and $x = 3y$?



→ Total area means negative area becomes positive.

Find points of intersection, Since we're in terms of y, Area = Right − Left

$3y = y^3 - y$

$0 = y^3 - 4y$

$0 = y(y^2 - 4)$

$y = 0, -2, 2$

$= 2\int_0^2 3y - (y^3 - y)\,dy$

$= 2\left[\frac{3y^2}{2} - \frac{y^4}{4} + \frac{y^2}{2}\right]_0^2 = \boxed{8}$

12. Find the derivative of $(\cos x)^{2x+5}$, do not simplify.

$y = (\cos x)^{2x+5}$

$\ln y = (2x+5)\ln(\cos x)$

e e

$$\boxed{y' = \left(2\ln(\cos x) + (2x+5)\left(-\frac{\sin x}{\cos x}\right)\right)e^{(2x+5)\ln(\cos x)}}$$

OR $\left(2\ln(\cos x) + (2x+5)(-\tan x)\right)(\cos x)^{2x+5}$

13. Find $\int 1 + \left(\frac{\cos^2 x}{\sin^2 x}\right) dx$

$= \int 1 + \cot^2 x\,dx = \int \csc^2 x\,dx$

$= \boxed{-\cot(x) + C}$